## A LINEARISATION USING PERTURBATION THEORY

In this appendix, we will flesh out the application of perturbation theory that leads from the Taylor series expansion in eq. (2.7) to the zeroth-order system of equations, eq. (2.8), and the first-order system of equations, eq. (2.9).

We start by substituting $U_i$ and $P$ in the governing RANS momentum equation, eq. (2.1), and the continuity equation, eq. (2.2), with the expansions from eq. (2.7) to yield the expanded momentum equation,

$$\begin{aligned}&\left(U_j^0+\varepsilon u_j^1+\varepsilon^2 u_j^2+\varepsilon^3 u_j^3+\dots\right)\frac{\partial}{\partial x_j}\left(U_i^0+\varepsilon u_i^1+\varepsilon^2 u_i^2+\varepsilon^3 u_i^3+\dots\right)\\&=\frac{\partial}{\partial x_j}K\left(\frac{\partial}{\partial x_j}\left(U_i^0+\varepsilon u_i^1+\varepsilon^2 u_i^2+\varepsilon^3 u_i^3+\dots\right)+\frac{\partial}{\partial x_i}\left(U_j^0+\varepsilon u_j^1+\varepsilon^2 u_j^2+\varepsilon^3 u_j^3+\dots\right)\right)\\&\quad-\frac{\partial}{\partial x_i}\left(P^0+\varepsilon p^1+\varepsilon^2 p^2+\varepsilon^3 p^3+\dots\right)+f_i,\end{aligned}\tag{A.1}$$

and the expanded continuity equation,

$$\frac{\partial}{\partial x_i}\left(U_i^0+\varepsilon u_i^1+\varepsilon^2 u_i^2+\varepsilon^3 u_i^3+\dots\right)=0.\tag{A.2}$$

The $n^{\text{th}}$-order equations are obtained by applying the operator $\frac{\partial^n}{\partial \varepsilon^n}$ to both sides and setting $\varepsilon=0$. To derive the zeroth-order equations then, we can start by applying the operator $\frac{\partial^0}{\partial \varepsilon^0}$ to equation (A.1), which does not change the equation, and then setting $\varepsilon=0$ to give

$$U_j^0\frac{\partial U_i^0}{\partial x_j}=\frac{\partial}{\partial x_j}K\left(\frac{\partial U_i^0}{\partial x_j}+\frac{\partial U_j^0}{\partial x_i}\right)-\frac{\partial P^0}{\partial x_i}.\tag{A.3}$$

The continuity equation and the boundary conditions, eq. (2.3), remain unchanged, leading to the result in eq. (2.8). Note that $U_i^{\text{lid}}$ is an externally prescribed quantity and therefore not a function of $\varepsilon$.

The first-order equations are derived by applying the operator $\frac{\partial}{\partial \varepsilon}$ to equation (A.1). Starting with the convective term, we can apply the product rule to find

$$\begin{aligned}&\frac{\partial}{\partial\varepsilon}\left[\left(U_j^0+\varepsilon u_j^1+\dots\right)\frac{\partial}{\partial x_j}\left(U_i^0+\varepsilon u_i^1+\dots\right)\right]\\&=\left(U_j^0+\varepsilon u_j^1+\dots\right)\frac{\partial}{\partial\varepsilon}\frac{\partial}{\partial x_j}\left(U_i^0+\varepsilon u_i^1+\dots\right)+\frac{\partial}{\partial x_j}\left(U_i^0+\varepsilon u_i^1+\dots\right)\frac{\partial}{\partial\varepsilon}\left(U_j^0+\varepsilon u_j^1\right)\\&=\left(U_j^0+\varepsilon u_j^1+\dots\right)\frac{\partial}{\partial x_j}\left(u_i^1+\dots\right)+\left(u_j^1+\dots\right)\frac{\partial}{\partial x_j}\left(U_i^0+\varepsilon u_i^1+\dots\right).\end{aligned}\tag{A.4}$$

Setting $\varepsilon=0$, you recover

$$\frac{\partial}{\partial\varepsilon}\left[\left(U_j^0+\varepsilon u_j^1+\dots\right)\frac{\partial}{\partial x_j}\left(U_i^0+\varepsilon u_i^1+\dots\right)\right]\bigg|_{\varepsilon=0}=U_j^0\frac{\partial u_i^1}{\partial x_j}+u_j^1\frac{\partial U_i^0}{\partial x_j}.\tag{A.5}$$

Taking the derivative with respect to $\varepsilon$ of the Reynolds stress term, and subsequently setting $\varepsilon=0$, we get

$$\begin{aligned}&\frac{\partial}{\partial\varepsilon}\left[\frac{\partial}{\partial x_j}K\left(\frac{\partial}{\partial x_j}\left(U_i^0+\varepsilon u_i^1+\dots\right)+\frac{\partial}{\partial x_i}\left(U_j^0+\varepsilon u_j^1+\dots\right)\right)\right]\bigg|_{\varepsilon=0}\\&=\frac{\partial}{\partial x_j}K\left(\frac{\partial}{\partial x_j}\left(u_i^1+\dots\right)+\frac{\partial}{\partial x_i}\left(u_j^1+\dots\right)\right)\bigg|_{\varepsilon=0}\\&=\frac{\partial}{\partial x_j}K\left(\frac{\partial u_i^1}{\partial x_j}+\frac{\partial u_j^1}{\partial x_i}\right).\end{aligned}\tag{A.6}$$

Following the same procedure with the pressure term, we obtain

$$\frac{\partial}{\partial\varepsilon}\left[-\frac{\partial}{\partial x_i}\left(P^0+\varepsilon p^1+\dots\right)\right]\bigg|_{\varepsilon=0}=-\frac{\partial}{\partial x_i}\left(p^1+\dots\right)\bigg|_{\varepsilon=0}=-\frac{\partial p^1}{\partial x_i}.\tag{A.7}$$

Finally, the forcing term is simply

$$\frac{\partial}{\partial \varepsilon}(\varepsilon f_i)\bigg|_{\varepsilon=0} = f_i. \tag{A.8}$$

Collecting these terms together, we arrive at the following first-order momentum equation

$$U_j^0 \frac{\partial u_i^1}{\partial x_j} + u_j^1 \frac{\partial U_i^0}{\partial x_j} = \frac{\partial}{\partial x_j} K \left( \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right) - \frac{\partial p^1}{\partial x_i} + f_i. \tag{A.9}$$

You will note that this is not the same as given in equation (2.9). We need to make two additional assumptions to arrive at equation (2.9), as set out at the end of section 2.2. We assume that the approaching flow is (i) along the $x$-axis, and (ii) only a function of the vertical coordinate, $z$, such that

$$U_i^0 = \delta_{i1} U^0(z) = \begin{bmatrix} U^0(z) \\ 0 \\ 0 \end{bmatrix}. \tag{A.10}$$

where $\delta_{i1}$ is the Kronecker delta function. This impacts only the terms on the left-hand side of (A.9). Substituting in the first term, we can replace $U_i^0$ with $\delta_{i1} U^0(z)$ to obtain

$$\begin{aligned} U_j^0 \frac{\partial u_i^1}{\partial x_j} &= \delta_{j1} U^0(z) \frac{\partial u_i^1}{\partial x_j} \\ &= \delta_{11} U^0(z) \frac{\partial u_i^1}{\partial x} + \delta_{12} U^0(z) \frac{\partial u_i^1}{\partial y} + \delta_{13} U^0(z) \frac{\partial u_i^1}{\partial z} \\ &= U^0(z) \frac{\partial u_i^1}{\partial x}. \end{aligned} \tag{A.11}$$

The second term is, analogously,

$$\begin{aligned} u_j^1 \frac{\partial U_i^0}{\partial x_j} &= u_j^1 \frac{\partial}{\partial x_j} \left( \delta_{i1} U^0(z) \right) \\ &= u^1 \frac{\partial}{\partial x} \delta_{i1} U^0(z) + v^1 \frac{\partial}{\partial y} \delta_{i1} U^0(z) + w^1 \frac{\partial}{\partial z} \delta_{i1} U^0(z) \\ &= w^1 \frac{\partial U^0}{\partial z} \delta_{i1}. \end{aligned} \tag{A.12}$$

Introducing eqs. (A.11) and (A.12) into eq. (A.7) now reflects eq. (2.9).

Finally, the first-order continuity equation is derived as

$$\frac{\partial}{\partial \varepsilon} \left( \frac{\partial}{\partial x_i} \left( U_i + \varepsilon u_i^1 + \ldots \right) \right) \bigg|_{\varepsilon=0} = \frac{\partial}{\partial x_i} \left( u_i^1 + \ldots \right) \bigg|_{\varepsilon=0} = \frac{\partial u_i^1}{\partial x_i} = 0, \tag{A.13}$$

and the boundary conditions are given as

$$\begin{aligned} \frac{\partial}{\partial \varepsilon} \left( U_i^0 + \varepsilon u_i^1 + \ldots \right) \bigg|_{\varepsilon=0} \bigg|_{z_0} &= 0 \\ \left( u_i^1 + \ldots \right) \big|_{\varepsilon=0} \big|_{z_0} &= 0 \\ u_i^1(z_0) &= 0, \end{aligned} \tag{A.14}$$

at roughness height $z_0$, and

$$\begin{aligned} \frac{\partial}{\partial \varepsilon} \left( U_i^0 + \varepsilon u_i^1 + \ldots \right) \bigg|_{\varepsilon=0} \bigg|_{z_i} &= \frac{\partial U_i^{\text{lid}}}{\partial \varepsilon} \\ \left( u_i^1 + \ldots \right) \big|_{\varepsilon=0} \big|_{z_i} &= 0 \\ u_i^1(z_i) &= 0 \end{aligned} \tag{A.15}$$

at inversion height $z_i$.